

P2 — Confounding diagnostic (Table 1)

Week 4 · applies L3 (Confounding) · optional and ungraded — no separate deadline, nothing to submit; M1 is assembled from these five increments Keep in your group GitHub repo: a reproduced Table 1 + a variable-role table + a crude-vs-adjusted comparison.

The question this week

How do your exposure groups differ, what causal role does each variable play, and what does that imply for adjustment? Table 1 is not a formatting exercise, and it is not where you decide what to control for. It describes how the groups differ and exposes sparse cells, positivity problems (a covariate level with almost nobody in one group), and coding errors; the adjustment set comes from causal and temporal reasoning about the variables it describes.

What to do

1. **Reproduce Table 1** stratified by your exposure, including the p-value column if the paper printed one — reproducing what they published is the point of this step. Then read the table for what it can actually show: how the groups differ, cells too sparse to support a model, positivity problems, and values that are impossible or miscoded. No Table 1 comparison selects your adjustment set, and a p-value least of all — it turns significant on sample size alone. Read the descriptive contrasts as magnitudes instead: the differences in means and proportions, or a standardized difference (the group gap in pooled SD units), which does not move with N.
2. **State the primary contrast** (which two groups, which exposure–outcome association) in one sentence.
3. **Classify the important variables** by role — **confounder** / **proxy** / **mediator** / **collider (including a pre-exposure collider)** / **instrument** / **outcome-only cause** — using temporal and causal reasoning (a small DAG helps). Prespecify a defensible confounder set. Note any candidate effect modifier separately: that is P3’s question, not a role in the DAG.
4. **Compare crude vs adjusted.** Fit an unadjusted association and one adjusted for your prespecified set, on the **same rows**, so any difference is attributable to adjustment. Report the move as what your adjustment set produces, and say what the comparison can and cannot establish: adjustment can remove confounding, but it can also remove real effect through a mediator or add bias through a collider, and this comparison cannot tell you which happened. If your measure is an odds or hazard ratio, it also moves through non-collapsibility, even with no confounding at all.

Deliverable

- Reproduced Table 1 (CSV + rendered) → your M1 **Table 1 slide**.
- A variable-role table (variable · role · reason · in adjustment set? y/n).
- Crude and adjusted estimates side by side, with a 2–3 sentence interpretation.

Worked example — MASLD/NHANES demonstration

- Continuous variables reproduce **exactly** (Age 51.0 (16.6), BMI 33.2, SBP/DBP identical) — a good sign your cohort is right.
- **Trap 1 — comorbidities are composites, not self-report.** Self-report alone under-counts: hypertension needs “self-report **or** measured BP $\geq 140/90$ ” to reach the paper’s 52.6% (54.1 reproduced); dyslipidaemia needs labs + meds to reach 57.7%.
- **Trap 2 — printed percentages that disagree with their own counts.** The income (PIR) percentages appear **interchanged**: the counts imply High 22.1% / Low 23.0%, but the paper prints 23.0 / 22.1. Catching arithmetic like this is exactly the P2 skill.
- **The age spread is the question, not the answer:** group ages run 40 → 62 years. The table reports the spread. Age is a confounder here for a causal and temporal reason, not a numeric one: it

raises phenotype membership *and* raises mortality, and it precedes the exposure — which is also what rules it out as a mediator.

What good looks like

- **Exceptional:** adjustment set justified from a DAG/temporal logic (not “everything”); notices a definitional or arithmetic problem like the PIR swap; crude-vs-adjusted change reported as what the adjustment set produces, without claiming which mechanism produced it.
- **Adequate:** Table 1 reproduced; roles mostly correct; sensible adjusted model.
- **Weak:** “adjusted for all variables in the dataset”; roles unassigned; no interpretation.

Be ready to defend (M3)

“Age differs hugely across your groups. Is it a confounder or is it on the causal path — and what would controlling for it do if you’re wrong?”

AI co-pilot note

A co-pilot can produce a fluent, well-formatted Table 1 that is **silently wrong** — in one tested reproduction, a **tableone** bug rendered *every* categorical comorbidity as **FALSE** (0 events) and no one noticed from the output alone. Spot-check counts against the raw variables.

Lab → project handoff

You are applying **L3 (Confounding)** — moving from Table 1 to crude, conditional, and marginal estimates (EpiMethods **confoundingE**). Justify the confounder set from science, not from a stepwise algorithm.